

Compiler and language-platform engineer focused on LLVM/C++, program analysis, and SSA. At MCST I implemented a LICM pass with conservative safety checks and a harness comparing behavior before and after optimization. Creator of UniversalToolchain/Wist2 with SSA, interpreter/CIL backends, and verifier-gated transformations; the system talk was accepted for LangDev 2026.

MCST · LLVM/C++

LICM pass: safety checks + semantic before/after regression harness

#1 of 49 · 104/104

PS-form memory-dependence analysis; only 5.0/5.0 score

LangDev 2026

Accepted talk on UniversalToolchain/Wist2 language architecture

EXPERIENCE		SKILLS
July 1 — August 31, 2026	MCST — compiler engineering intern - Implemented an LLVM LICM pass in C++ with conservative checks for side effects, memory access, loop invariance, preheaders, and speculative safety. - Built a Python harness with baseline-vs-optimized execution through lli, opt -verify-each, and explicit CHANGE/NO_CHANGE expectations.	Compiler / LLVM C++23, LLVM passes, LoopInfo, CFG, loop invariance, LICM, SSA, dominance Program analysis memory dependence, dataflow, exact oracles, differential/metamorphic/randomized testing IR / runtime .NET, Bytecode/AIR, SSA, interpreter, CIL backend, verifier-gated transformations Tooling Python harnesses, CMake, Linux, clang-tidy, ASan/UBSan, Git
2026	PS-form Memory Dependence Analyzer — #1 of 49 - Implemented conservative analysis of parametric memory accesses; passed 104/104 official tests and received the only 5.0/5.0 score. - Built an exact oracle for small domains plus metamorphic/randomized testing for reproducible counterexamples.	
2024 — present	UniversalToolchain / Wist2 — creator and primary developer - Built a modular compiler/runtime platform with Bytecode/AIR, SSA, interpreter, and CIL backend. - Develop verifier-gated transformations and interpreter/CIL parity with fail-closed compiler-contract checks.	EDUCATION Software Engineering student at HSE University RECOGNITION The LangDev 2026 Program Committee accepted a talk on the architecture of UniversalToolchain/Wist2 for presentation in Torremolinos, Spain. Moscow / remote
PROJECTS		
UniversalToolchain / Wist2 The current exact test manifest passes with zero failures; interpreter/CIL parity, clean-consumer scenarios, and reproducible compiler/runtime contracts are verified. PS-form Memory Dependence Analyzer Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples. x86-64 Codegen & Register Allocation Lab A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.		